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Growth of environmentally stable transition metal selenide films

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Two-dimensional transition metal selenides (TMSs) possess fascinating physical properties. However, many as-prepared TMSs are environmentally unstable and limited in sample size, which greatly hinder their wide applications in high-performance electrical devices. Here we develop a general two-step vapour deposition method and successfully grow different TMS films with controllable thickness, wafer size and high quality. The superconductivity of the grown $NbSe_2$ film is comparable with sheets exfoliated from bulk materials, and can maintain stability after a variety of harsh treatments, which are ascribed to the absence of oxygen during the whole growth process. Such environmental stability can greatly simplify the fabrication procedure for device applications, and should be of both fundamental and technological significance in developing TMS-based devices.

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Two-dimensional (2D) materials possess excellent electrical, mechanical, thermal and optical properties. Among them, 2D transition metal selenides (TMSs) hold great promise for future applications in low-power, high-performance and flexible electrical devices[1], which have attracted increasing attentions. For instance, 2D TMSs have been considered to be ideal platforms for exploring the novel physical properties of optoelectronics[2], charge density waves (CDWs)[3–5] and superconductivity[6–8] in low dimensions. However, most TMSs are prone to react with H_2O or O_2 in the environment and are stable only in inert gas or vacuum, or beneath a protective layer. Therefore, maintaining the stability of TMSs, tellurides, halides and other 2D materials remains a huge technological challenge[3–10]. It greatly hinders the characterization, performance testing, device fabrication and potential application of these 2D materials.

Similar to graphene, most TMSs can be prepared by top-down or bottom-up approaches[11]. Types of TMS domains have been grown by molecular beam epitaxy[12–14], with high electrical quality in local regions. However, those grown domains are unstable when exposed to the environment without any capping layer, thus hindering further applications in electrical devices. Among recent preparation methods, chemical vapour deposition (CVD), a compatible technique with the semiconductor industry, has been successfully utilized to grow a variety of 2D materials, including graphene[15], hexagonal boron nitride[16] and TMSs[17–21]. Wafer-sized sulphide films have already been grown by a developed metal organic CVD method[19], and dozens of micrometre-sized transition metal chalcogenide domains have been produced recently by molten salt-assisted methods[20–22]. However, TMSs and some tellurides show poor environmental stability and their intrinsic properties (for example, high-temperature superconductivity and non-saturating magnetoresistance) will vanish rapidly even after exposure to air for a short time, which seriously limits their potential application. Until now, the preparation of superconducting selenide films with high environmental stability still remains challenging.

During CVD growth, oxygen inevitably exists in the atmosphere or precursors, so oxygen bonds to TMSs will be easily formed even in the reaction chamber. Therefore, a total oxygen-free condition should be the precondition for growing environmentally stable TMSs. Here, we successfully develop a two-step vapour deposition method to grow stable TMS films by combining the typical physical vapour deposition (PVD) and CVD methods, in which the H_2O/O_2 is completely removed from both processes. The samples can be homogeneously grown to wafer size or in patterns with shadow masks. Although similar growth methods have been proposed[18,23–27], their sample crystallinity is not comparable with those produced by the traditional CVD[17] or molten salt-assisted CVD methods[21]. To obtain TMS films with high crystallinity, we conduct magnetron sputtering in a H_2O/O_2-free condition by deep degassing to deposit the metal films on a heated sapphire substrate, instead of the widely used electron-beam or thermal evaporation on SiO_2/Si wafers[18,25]. Subsequently, a typical CVD process is applied to transform the crystalline metal films into 2D TMSs, halides or carbides with different source materials. Here we take $NbSe_2$ films, a typical 2D superconductor, as an example to investigate the structural stability and the corresponding superconductivity of 2D TMSs grown by our two-step method.

Figure 1a schematically illustrates the structural transformation from the Nb film to the layered $NbSe_2$ via selenization. Figure 1b shows the as-deposited Nb and $NbSe_2$ films on 2-inch double-side polished sapphire wafers, which are macroscopically homogeneous and transparent. The smoothness and thickness of these films are measured *ex situ* by atomic force microscopy (AFM), as shown in Fig. 1c. The thickness of the sputtered Nb film is measured to be ~1.6 nm, which corresponds to an actual film thickness of ~0.7 nm, considering the surface distance between the Nb film and the sapphire substrate[28] (see details in Supplementary Fig. 1); in the meantime, the surface roughness is measured to be ±0.25 nm, similar to that of the sapphire substrate underneath. After selenization, the Nb film is transformed to the $NbSe_2$ film, accompanied by a thickness increase to ~2.3 nm (as measured by AFM in Fig. 1c), which corresponds to an actual film thickness of ~1.5 nm (Supplementary Fig. 1). This huge change in film thickness (~100% increment) implies the occurrence of structural transformation in both the interlayer

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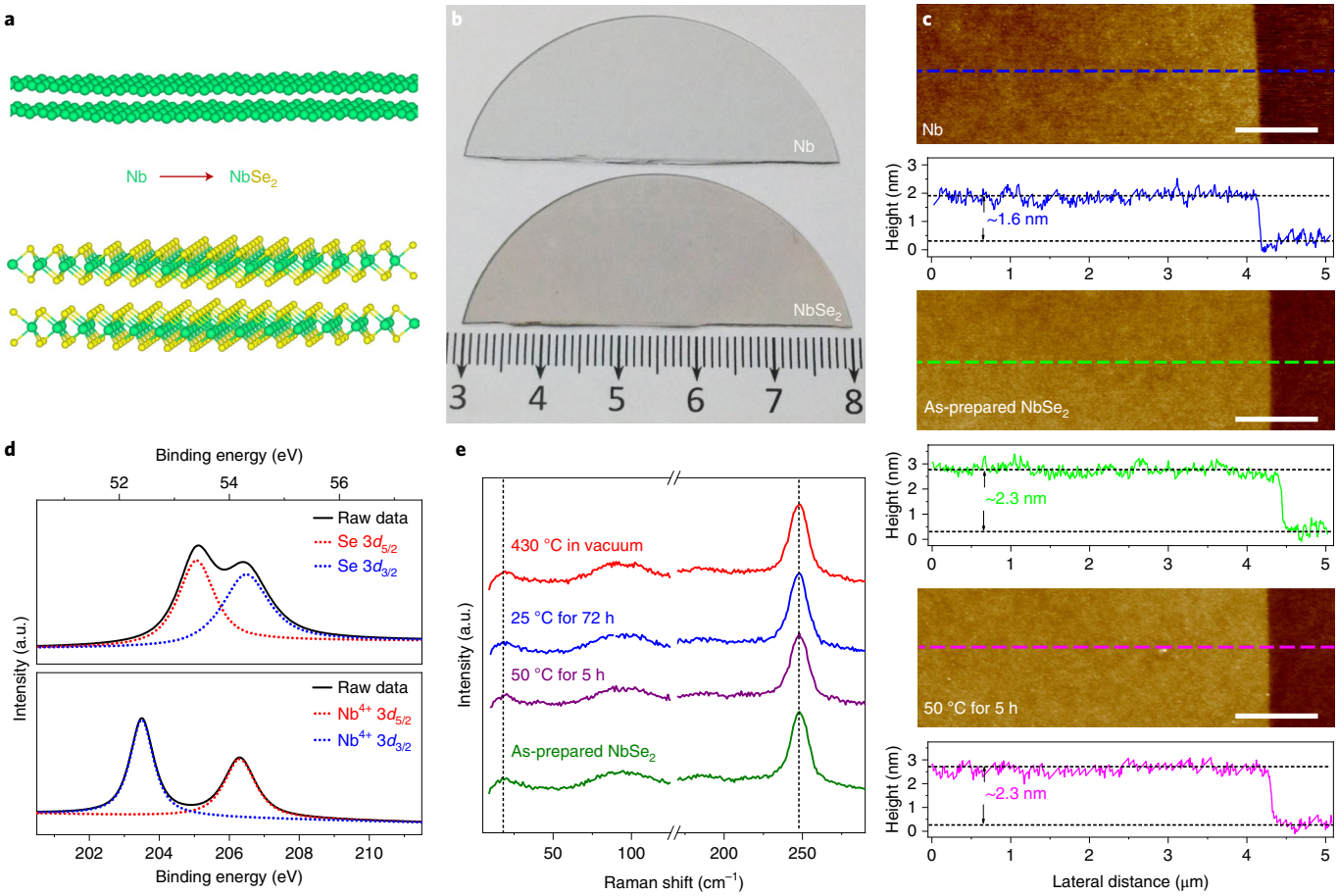

Fig. 1 | Two-step vapour deposition growth of wafer-sized $NbSe_2$ film. a, A schematic illustration of the structural transformation from the sputtered Nb film to a 2*H* $NbSe_2$ film. **b**, A photograph of a Nb film and a selenized Nb film on the 2-inch double-side polished sapphire wafers, which are homogeneous and transparent. **c**, AFM images and the corresponding height profiles of an as-sputtered Nb film, a selenized ($NbSe_2$) film and a heat-treated (50 °C for 5 h in air) $NbSe_2$ film. A height increase of ~50% is exhibited after selenization. Scale bars, 1 μm. **d**, XPS spectra of Nb 3*d* and Se 3*d* from a selenized Nb film (0.7 nm in thickness) after long-time exposure to air. **e**, Raman spectra with perpendicular polarization for the $NbSe_2$ films after different treatments.

and intralayer structures during the selenization. More importantly, the thickness and roughness of the films remain nearly constant even after further heat treatment (for example, heating at 50 °C in air, Supplementary Fig. 2), indicating that the structure and morphology of the grown $NbSe_2$ films are environmentally stable.

These selenized films are further characterized by X-ray photoelectron spectroscopy (XPS) and Raman spectroscopy. According to the XPS measurements in Fig. 1d, the binding energies for the Se $3d_{5/2}$ and $3d_{3/2}$ doublets are 53.2 eV and 54.2 eV, and those for the Nb^{4+} $3d_{5/2}$ and $3d_{3/2}$ are 203.4 eV and 206.3 eV, respectively. These results are consistent with the corresponding binding energies of Se and Nb reported in bulk $NbSe_2$ and CVD-grown few-layer $NbSe_2$ sheets[20]. The absence of a prominent peak at ~202 eV (corresponding to the Nb^0 $3d_{5/2}$, see the survey scan in Supplementary Fig. 3) indicates the complete selenization of the Nb film. A terahertz Raman spectrum with perpendicular polarization is further used to measure the low-frequency mode of a ~1.5-nm-thick $NbSe_2$ film (Fig. 1e), where a peak at ~20 cm^{-1} is observed, corresponding to the shear mode frequency. This indicates the dominance of the bilayer $NbSe_2$ with high crystallinity in the film[4]. The Raman spectra with parallel polarization of $NbSe_2$ films have two characteristic peaks located at ~226 cm^{-1} and ~247 cm^{-1}, which are assigned to A_{1g} and E_g[29], respectively (Supplementary Fig. 4). After vacuum annealing, long-time exposure and even heating in air, no obvious changes in the peak positions and relative intensities are detected in these $NbSe_2$ films. Therefore, the $NbSe_2$ films grown by this two-step vapour deposition method exhibit large-scale homogeneity (Supplementary Fig. 5) and high environmental stability. We further examine the amount of residual Se by comparing the Raman spectra between selenized films and pure Se (Supplementary Fig. 6). Clearly, no obvious Raman peak from Se is detected in the properly selenized film, while the peaks from residual Se can be detected if the film is excessively selenized. It is also noticed that the residual Se usually agglomerates into nanoparticles on the surface of $NbSe_2$ films, rather than forming a protective Se layer.

The grown 1.5 nm $NbSe_2$ films without any protective layers are further characterized by high-angle annular dark-field scanning transmission electron microscopy (STEM) after the traditional wet transfer[15]. The energy-dispersive X-ray spectrometry mapping confirms that the Nb and Se are homogeneously distributed in the film over a large region (Supplementary Fig. 7). The STEM image in Fig. 2a shows that the $NbSe_2$ film has a polycrystalline structure with an average lateral grain size of ~50 nm. The zoomed-in image (Fig. 2b) of the neighbouring grains with a misorientation of ~25° shows that both grains have the same stacking structure, and the grain boundary is mostly composed of 5–7 pairs, without any octagon. These 5–7 pairs will interlink across the van der Waals interlayers through the dangling bonds, which leads to the interlayer coupling at the grain boundary region. Figure 2c shows a typical STEM image inside an individual grain (confirmed by the inset fast Fourier transform pattern), demonstrating a near-perfect hexagonal lattice across the 25 nm × 25 nm region. The zoomed-in image

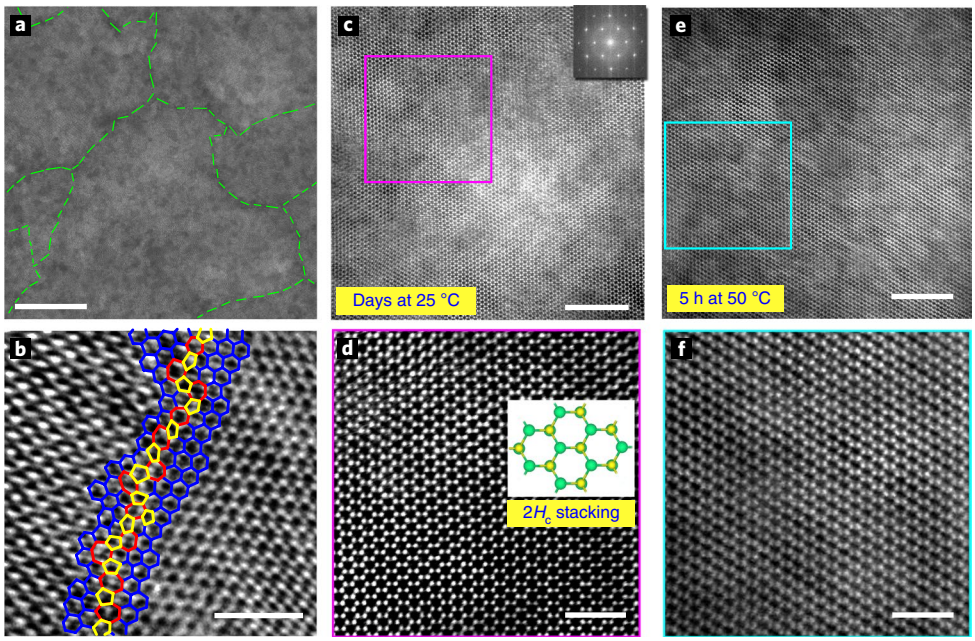

Fig. 2 | Atomic images of a 1.5 nm $NbSe_2$ film. a, A low-magnification STEM image showing the polycrystalline structure of a $NbSe_2$ film; the grain boundaries are marked by green dashed lines. **b**, The atomic structure of a tilt grain boundary with a misorientation of ~25°, where the pentagon (yellow), heptagon (red) and hexagon (blue) structures in the grain boundary region are marked out accordingly. Only 5–7 pairs exist. **c**, The lattice structure in a single-crystalline grain, showing a perfect single-crystalline structure (as confirmed by the inset fast Fourier transform pattern), where no obvious atomic defect is detected even after exposure to air for several days. **d**, Atomic structure image of the area marked out by a purple square in **c**, with the inset showing the atomic model of $NbSe_2$ (top view) with 2H_c structure. **e**,**f**, STEM images at different magnifications showing the atomic structure of the $NbSe_2$ film after heat treatment at 50 °C for 5 h in air, where no obvious atomic vacancy is observed. Scale bars, 20 nm (**a**), 2 nm (**b**,**d**,**f**), 5 nm (**c**,**e**).

of the area marked out by the purple square further confirms the absence of atomic vacancy in the $NbSe_2$ film (Fig. 2d). Considering the prevalence of lattice defects in CVD-grown $NbSe_2$ grains or mechanically exfoliated monolayer materials[20,30], these grown $NbSe_2$ films with near-perfect crystalline structure should have improved structural stability for device applications.

Since this $NbSe_2$ film with homogeneous thickness of ~1.5 nm is mainly bilayer, the atomic image in Fig. 2d should correspond to a homogeneous 2H_c stacking structure[20], as schematically illustrated in the inset. Subsequently, this $NbSe_2$ film transferred onto a TEM grid is further exposed to the atmosphere at 50 °C for 5 h to test its structural stability. Surprisingly, the homogeneous 2H_c stacking structure is perfectly retained after the heat treatment, without any observable damaged or obvious atomic vacancy in the film, as demonstrated by the atomic STEM images in Fig. 2e,f. All of these results indicate a robust atomic configuration in the bilayer $NbSe_2$ films grown by the two-step method.

We investigate the environmental stability of $NbSe_2$ film by measuring its superconductivity after different treatments. According to previous reports, most 2D selenides lack environmental stability[21]. For example, the superconductivity of $NbSe_2$ film grown by a molten salt-assisted CVD vanishes quickly once exposed to air (even when protected by graphene layers)[20]. Here, we first ensure that the sputtered 3 nm Nb film shows no superconductivity (Supplementary Fig. 8) and the bilayer $NbSe_2$ film (without any protective layer) shows a typical metallic conductivity with a superconducting transition critical temperature (T_c, the temperature where the sheet resistance drops to 10% of its normal state) at 2.77 K. Owing to the polycrystalline structure of the $NbSe_2$ films, the T_c is increased to 2.89 K when the linewidth is narrowed down from 500 μm to 5 μm (Supplementary Fig. 9). The voltage–current characteristics at selected temperatures follow the relation of $V \propto I^{\alpha}$ (with α being the exponent deduced from the power-law behaviour), as predicted by the Berezinskii–Kosterlitz–Thouless (BKT) model[31], and T_{BKT} is fitted to be 2.81 K when α reaches 3 (Supplementary Fig. 10). This value is consistent with the T_c, indicating the occurrence of a BKT transition. The normalized resistances R/R_{10K} as a function of temperature for $NbSe_2$ films with thicknesses ranging from 1.5 nm to 11.2 nm are shown in the right inset of Fig. 3a. It is clearly seen that T_c shows a strong dependence on the thickness of the $NbSe_2$ films, where a lower T_c is exhibited in the thinner films. The influences of external out-of-plane and in-plane magnetic fields on the superconducting characteristics are shown in Fig. 3b and Supplementary Fig. 11. The temperature dependences of H_{c2} under out-of-plane and in-plane magnetic fields are plotted and presented in the inset of Fig. 3b. As described by the Ginzburg–Landau theory[32] of 2D superconductors, the coherence length and effective thickness of our $NbSe_2$ film can be fitted to be ~7.8 nm and ~2.6 nm, respectively (see Supplementary Fig. 11). Given the smaller value of effective film thickness than the coherence length, the superconducting phase transition occurring in $NbSe_2$ film should be of the BKT type, further confirming that the grown $NbSe_2$ films are true 2D superconductors. The critical fields (H_{c2}) at 0 K in the parallel and perpendicular magnetic fields are fitted to be ~52.1 T and ~5.4 T (inset of Fig. 3b), respectively, which are much larger than those of micromechanically cleaved $NbSe_2$ sheets[6]. This enhancement of critical field should originate from the interlayer coupling at the grain boundary regions (Fig. 2b).

Next, we further study the environmental stability of the bilayer $NbSe_2$ films by tracing their T_c changes after different treatments. Figure 3c shows the T_c evolution of $NbSe_2$ films with exposure time in air (see details in Supplementary Fig. 12). After exposing at 25 °C for 1 day, the T_c remains nearly unchanged. When the exposure time is extended to 10 days, the reduction in T_c remains negligible (less than 0.1 K); more interestingly, the $NbSe_2$ films still show robust

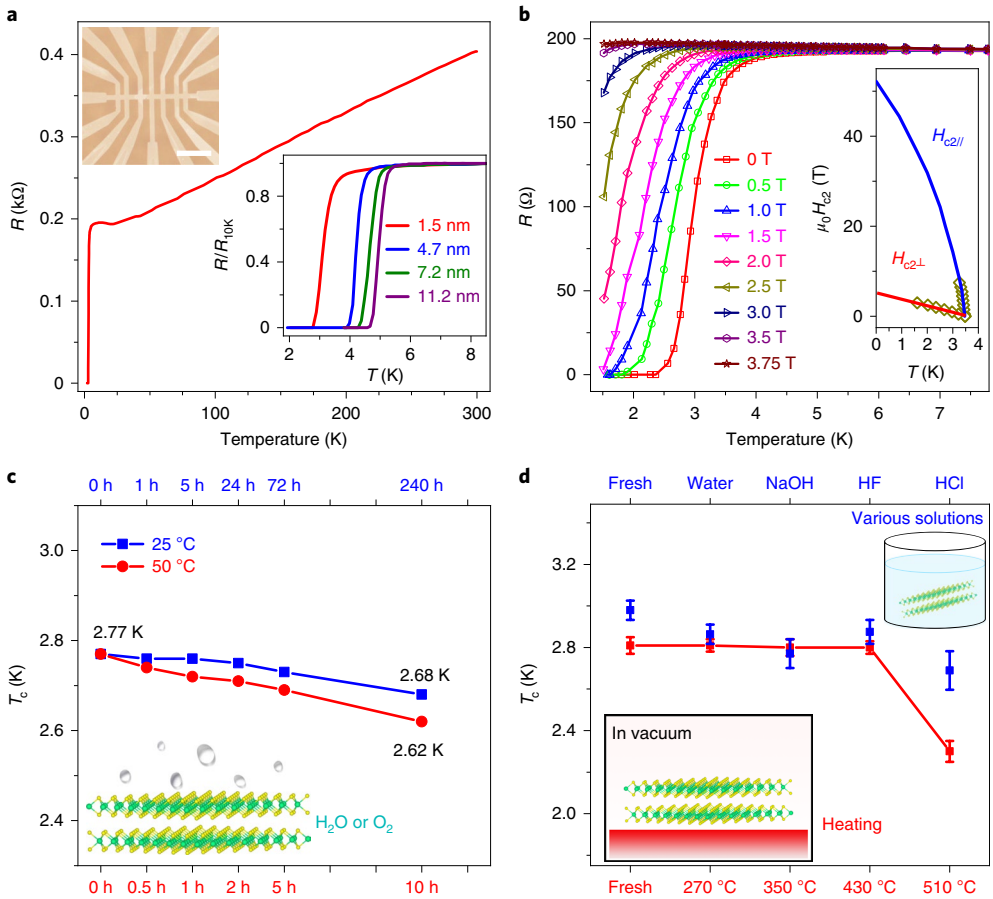

Fig. 3 | Superconductivity in $NbSe_2$ films with different treatments. a, The temperature dependence of the longitudinal resistance for bilayer $NbSe_2$ films. The top left inset shows a typical Hall bar device by pattern growth (scale bar, 500 μm); the bottom right inset shows the detailed superconductivity in 1.5 nm, 4.7 nm, 7.2 nm and 11.2 nm $NbSe_2$ films, respectively. **b**, The superconducting resistance transition of the bilayer film under an out-of-plane magnetic field. The inset shows the temperature dependence of the upper critical field H_{c2} under out-of-plane (red line) and in-plane (blue line) magnetic fields, and the solid line is the linear fitting. **c**, The time-dependent evolution of superconducting T_c in bilayer $NbSe_2$ films after exposure to air at different temperatures, illustrated by the inset image. The blue and red lines correspond to the exposures at 25 °C and 50 °C, respectively. **d**, Robust superconductivity of bilayer $NbSe_2$ films after vacuum annealing for 60 min (red line) and being immersed in different aqueous solutions for 30 min (blue error bars), illustrated by the bottom and upper inset images, respectively. All of the T_c error bars represent the standard deviation of the results obtained from independent superconductivity measurements.

superconducting behaviour even after heating at 50 °C for 10 h, with a reduction of T_c of merely 0.15 K (Fig. 3c and Supplementary Fig. 13). Additional harsh treatments are also performed, such as immersion of the as-prepared films in different aqueous solutions for 30 min (including deionized water, 0.1 M NaOH solution, 8% HF solution and 0.1 M HCl solution), with the corresponding changes in T_c shown in Fig. 3d. Clearly, the T_c values of the $NbSe_2$ films show nearly no change after immersion of the films in deionized water or HF solutions, and only a slight reduction (less than 0.21 K) is observed after immersion in NaOH or HCl solutions. With such incredible environmental stability, the structure and morphology of $NbSe_2$ films can be well maintained even after the traditional wet transfer (Supplementary Fig. 14). To test the thermal stability of the $NbSe_2$ films, annealing is also performed in high vacuum ($<10^{-5}$ Pa) for 60 min, followed by exposure to atmosphere for more than 60 min. Figure 3d shows that the corresponding T_c values can be effectively retained until the annealing temperature exceeds 430 °C. Thus, the above tests unambiguously confirm the excellent environmental and thermal stability of $NbSe_2$ films produced by our two-step vapour deposition method, which holds significant implications for the future applications of TMSs in flexible electrical devices.

For comparison, we pre-oxidize the Nb film at 400 °C in air before selenization (similar to the metal oxide precursors in previous studies, as shown in Supplementary Fig. 15). The T_c of as-prepared $NbSe_2$ films from pre-oxidized Nb films is only 1.88 K, which is nearly 1 K lower than that of $NbSe_2$ films grown from the pure Nb films. After further exposure in the atmosphere for 1 day, the electrical resistance of this $NbSe_2$ film shows a significant increase by 50%, and no superconductivity can be detected. Since oxygen is completely excluded from the growth processes and the precursors in our two-step vapour deposition, we propose that the superior structural stability of our $NbSe_2$ films originates from the absence of atomic vacancy and oxygen bonds, which often cause the activated reactive sites in the grown $NbSe_2$ films and thereby result in poor structural stability of TMSs[30]. Moreover, their structural stability can be largely impaired after Ar ion irradiations, which can easily introduce atomic vacancies into most 2D materials (Supplementary Fig. 16)[33]. This hypothesis is further confirmed by the theoretical simulation of oxidization of few-layer 2*H* stacked $NbSe_2$, as shown in Supplementary Fig. 17. The oxygen can be easily trapped around the pre-existing oxygen bonds on the $NbSe_2$ surface, which could then diffuse easily into the interlayers, as facilitated by the vacancies, resulting in a poor structural stability of TMSs.

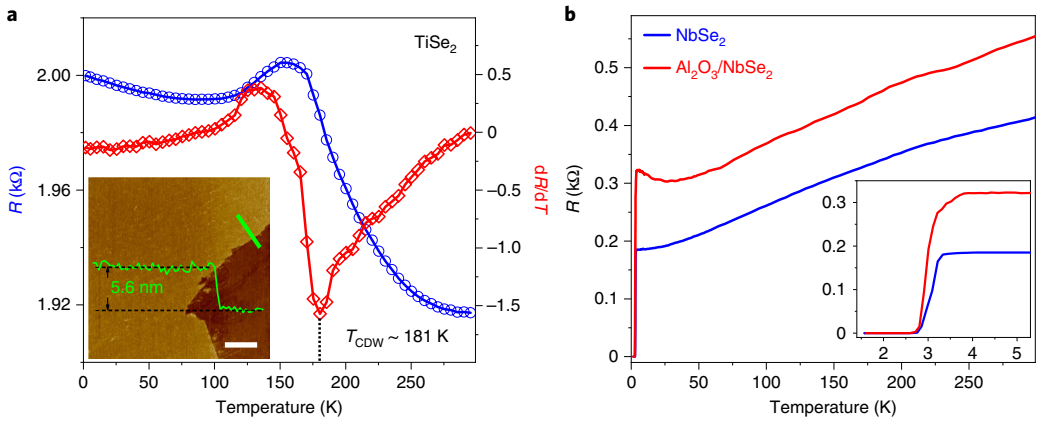

Fig. 4 | Generalization of the two-step vapour deposition method. a, Temperature dependence of the resistance of a 5.6 nm $TiSe_2$ film. The inset shows the corresponding AFM morphology. Scale bar, 2 μm. **b**, Superconductivity of an as-prepared bilayer $NbSe_2$ film and a bilayer $NbSe_2$ encapsulated with a 5-nm-thick Al_2O_3 layer by thermal evaporation. The inset shows the detailed evolution of the electrical resistance.

Once deprived of oxygen in the whole growth process, the selenides can exhibit high environmental stability. This mechanism is not only applicable to the growth of $NbSe_2$ film, but also to the 1*T* $TiSe_2$, as demonstrated by the controlled growth of $TiSe_2$ films using our two-step vapour deposition method (Fig. 4a). $TiSe_2$ is known to have exciting properties of superconductivity (with heavy doping) and CDW at low dimensions[5,34]; however, its environmental stability is rather poor. Here, we examine the stability of our grown $TiSe_2$ films. Figure 4a shows a typical *R*–*T* curve of the 5.6-nm-thick $TiSe_2$ film, which is exposed to the atmosphere for a few days before measurement (see details in Supplementary Fig. 18). An obvious inflection still appears in the temperatures ranging from 150 K to 200 K. The minimum value of d*R*/d*T* indicates that the T_{CDW} of this film is ~181 K, which is similar to that of the $TiSe_2$ sheets exfoliated from bulk material[34]; nevertheless, our film exhibits a much better environmental stability.

An important application of 2D TMSs is the fabrication of stacking structure[1], or Josephson junctions[35] in 2D superconductors. These processes usually require stacking different 2D materials with an ultrathin isolator layer, which probably destroys the 2D materials during fabrication. To test the stability of our $NbSe_2$ films in this application, we cover the as-grown 1.5 nm $NbSe_2$ film with a 5 nm aluminium oxide (produced by the oxidization of thermally evaporated aluminium in air). After the destructive treatment, the T_c of the $NbSe_2$ film remains nearly unchanged except a slight reduction in conductivity (Fig. 4b). On full contact with the oxygen bonds of the capping aluminium oxide layer, a fluctuation of electrical resistance in the bilayer $NbSe_2$ appears at ~26 K, which indicates a limited effect of this destructive encapsulation process on the properties of the $NbSe_2$ film.

In summary, we successfully develop a general two-step vapour deposition method and grow different 2D TMSs with superior structural stability, including $NbSe_2$ and $TiSe_2$ films. These TMS films can grow up to wafer size and possess a homogeneous structure, as well as high quality. Both the harsh treatments and destructive encapsulation experiments demonstrate that the TMS films fabricated by our two-step method possess excellent environmental stability and superior physical properties. Such high environmental and structural stability of the grown TMSs allows the easy stacking with other materials into a superlattice, thereby greatly simplifying the fabrication of TMS-based devices. Therefore, the growth of environmentally stable TMS films is of both fundamental and technological significance to the development of TMS-based devices with complicated integration structure.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at https://doi.org/10.1038/s41563-019-0321-8.

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Author contributions

L.G. conceived and supervised the project, and designed the experiments; H.L. carried out growth experiments and transport measurements; J.X., X.H. and W.S. helped with the device fabrication and transport measurements; Q.Z. and J.W. performed STEM; D.L. and X.X. performed the terahertz Raman spectroscopy; D.S. carried out the theoretical simulations; L.G. and H.L. wrote the manuscript, J.W., Q.Z., J.X. and X.X. revised it, and all of the authors commented on it.

Competing interests

The authors declare no competing interests.

Additional information

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Methods

Growth of selenide films. We adopt a two-step vapour deposition to grow the selenide films, which includes the typical PVD (magnetron sputter) and CVD processes. The typical process for the individual steps is as follows.

For PVD, in this study, we adopt the magnetron sputter to deposit the metal films, which consists of a main sputter chamber (base pressure $<10^{-4}$ Pa) and a loadlock chamber (base pressure $< 10^{-1}$ Pa). The loadlock chamber is crucial for degassing the main chamber. If the main chamber is exposed to the atmosphere, it should be degassed twice by pre-heating in a vacuum (>500 °C, $<10^{-4}$ Pa for more than 30 min) at first, followed by pre-sputtering of the metal target (twice). Afterwards, the substrates can be loaded into the main chamber through the loadlock chamber for deposition. All of the high-purity (>99.95 at.%) metal targets are commercial, which need to be pre-sputtered to eliminate surface contamination before the sputtering process. The standard sputtering is conducted under the following conditions: substrate temperature, ~300 °C; deposition rate, ~0.2 Å s^{-1}; and chamber pressure, 10 Pa. After the sputtering, the as-obtained metal films are stored in a glovebox before further experiments.

For CVD, in this study, we adopt a two-zone tube furnace (base pressure $<10^{-1}$ Pa) for the selenization. Before the selenization process, sufficient baking of the quartz tube (>600 °C, <1 Pa, >30 min) is carried out. After loading the metal films, thrice pumping to vacuum (<1 Pa) and inflating with Ar or H_2 to atmospheric pressure are needed to further remove the residual H_2O/O_2 in the tube. Subsequently, the typical selenization process is conducted in the pre-baked and degassed two-zone furnace under atmospheric pressure, where Se (>99.9%) is placed in zone I (280 °C) upstream and the Nb films are heated to 600 °C for 30 min in zone II downstream. During the selenization, a gas mixture of H_2/Ar (1:10) is used as the carrier gas and a typical growth time for a homogeneous film in our study is 8 min. Finally, the grown films are annealed at 350 °C for 30 min in zone II to remove the redundant Se (zone I is not heated at this stage). A similar two-step process is employed to grow the $TiSe_2$ films, in which the zone-I and zone-II temperatures are adjusted to 400 °C and 650 °C, respectively. The grown films are transferred by the traditional wet method to different substrates or TEM grids for further characterization.

Characterization. STEM images are captured by a FEI Titan ChemiSTEM with a probe aberration corrector, which is operated at 200 kV to minimize the knock-on damage. Raman spectra are acquired using a Witec/alpha 300 R confocal microscope with a 532 nm laser at atmospheric conditions, and polarization-dependent terahertz Raman analyses (also at 532 nm) are performed in vacuum with a grating spectrometer equipped with a liquid-nitrogen-cooled CCD (charged coupled device) from Princeton Instruments. The laser powers are lower than 1 mW during the measurements. AFM images are taken using the Bruker Dimension Icon in tapping mode. XPS analyses are performed on a Scienta omicron DA-30 electron spectrometer using Al K_α radiation ($h\nu = 1486.6$ eV), and the peaks are fitted by the approximation of Lorentzian–Gaussian mixed function.

Electrical measurements. The superconducting and transport measurements are carried out in a ^{4}He cryostat with a superconducting magnet (Oxford Teslatron 8 T), and electrical transport measurements are performed using a standard lock-in amplifier (Stanford SR830) with currents of 1~10 μA. For all measurements if no specified, four-probe contacts are used by an electron-beam-evaporated 80 nm Cr/Au array (200 μm dots with 500 μm spacing). For the transport measurements with narrower linewidth, the grown films are also fabricated into a Hall bar pattern (~5 μm linewidth) by standard electron beam lithography and reactive ion beam etching. All stability comparisons by T_c are performed three times, and we use their average values unless specified otherwise.

Data availability

The data that support the findings of this study are available from the corresponding author upon request.